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ORIGINAL ARTICLE

EPIDEMIOLOGY AND CLINICAL MEDICINE

Plasma myoglobin indicates muscle damage associated with acceleration/deceleration during football

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ABSTRACT

BACKGROUND: Monitoring muscle damage in athletes assists not only coaches to adjust the training workload but also medical staff to prevent injury. Measuring blood myoglobin concentration can help evaluate muscle damage. The novel portable device utilized in this study allows for easy on-site measurement of myoglobin, providing real-time data on the player's muscle damage. This study investigated the relationship between external load (global positioning system parameters) and internal loads (myoglobin concentration and creatine kinase activity) in 15 male professional football players before and after a match.

METHODS: Whole blood samples from participants' fingertips were collected before the match (baseline) and at 2, 16, and 40 h after the match. Myoglobin concentrations were measured using the IA-100 compact immunoassay system. Creatine kinase concentrations were measured in a clinical laboratory, and match loads were monitored using a global positioning system device.

RESULTS: The mean myoglobin concentration was significantly higher at 2 h than at the other time points ($P<0.05$), and decreased to baseline levels within 16 h post-match. The mean creatine kinase concentration increased after the match but did not reach a significant level. Muscle damage monitored by myoglobin after football match-play was strongly associated with acceleration/deceleration metrics rather than the sprint/high-speed running distance.

CONCLUSIONS: Our findings indicate that myoglobin is a more sensitive marker of muscle damage than creatine kinase after football match-play. Monitoring myoglobin in athletes can aid in determining their recovery status from the previous training load and help practitioners manage the training load.

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KEY WORDS: Myoglobin; Running; Geographic information system; Soccer; Creatine kinase.

Plasma myoglobin (MB) concentration and creatine kinase (CK) activity are primary diagnostic biomarkers for estimating muscle damage in patients with myocardial infarction and athletes with exercise-induced muscle damage.[1] MB measurements are preferred in emergencies, such as when initially diagnosing myocardial infarction, as they are more sensitive than CK activity measurements.[2] However, CK activity is commonly used to monitor athletes for exercise-induced skeletal muscle damage as the test for CK activity is convenient and inexpensive.[3]

MB is a cytoplasmic hemoprotein expressed in cardiac myocytes and skeletal muscle, and is involved in oxygen storage and diffusion.[4] Furthermore, MB is released into the blood from muscle tissues after muscle damage.[5] CK,

which is an enzyme, is primarily found in the heart and skeletal muscle, and phosphorylates creatine.[6] Therefore, CK is important for muscle energy homeostasis.[6] Similar to MB, plasma CK concentrations increase after muscle damage.[7] Other pathological conditions, such as rhabdomyolysis and myasthenia gravis, also increase the concentrations of these muscle-derived biomarkers.[7]

Monitoring an athlete's external load using wearable devices, such as a global positioning system (GPS), during a football match can help avoid injury by individualizing the player's training load the following week using the GPS data.[8] In addition, an athlete's internal load, such as muscular, neuromuscular, and mental fatigue,[9] contributes to the player's athletic health. Muscle damage markers, such as CK and the countermovement jump, are commonly used to estimate the athlete's internal load.[9] All people involved with the players, such as the head coach, fitness coach, medical staff, and nutritionist, usually share the external and internal monitoring data to assess players' training effects.[10] Therefore, including scientific-based evidence in this assessment would be beneficial.

Measuring MB in patients with heart disease is common in the hospital setting, but uncommon in sports for athletes with muscle issues because there has been no convenient device to measure MB concentrations.[9] However, a novel portable immunoassay device called the IA-100 (Sysmex Corporation, Kobe-city, Hyogo, Japan) measures MB concentrations from a finger prick (35 µL of capillary blood) and may change this difficulty in measurement. Within 20 minutes of measurement, athletes can obtain their MB concentration by accessing the data using a smartphone application. Therefore, this device allows athletes to conveniently monitor their MB concentration without going to a clinic.

This study investigated the relationship between external (GPS parameters) and internal (MB concentration and CK activity) loads in professional football players before and after a match. We hypothesized that the MB concentration, measured using the IA-100, is a more sensitive biomarker of muscle damage after a match than CK activity.

Materials and methods

Participants

The study participants included 15 male professional football players with an average age of 24.3 years who played in the 2021 season for the IWAKI FC (Japan Football League, Fukushima, Japan). The ethics committee of the university (E21-0008-M01) approved the study protocol before initiation of the study. The study has been conducted in accordance with the principles set forth in the Helsinki Declaration. Written informed consent was obtained from the players who participated in this study.

Observational approach

This observational study was conducted during the in-season Japan Football League official 31st match with a total of 34 sections. Of the team members, 18 players, including two goalkeepers, were selected according to the match statistics, and 15 field players played this match. Six players played full-time (90 min), two players played 80 min, one player played 70 min, one player played 60 min, one player played 30 min, two players played 20 min, and two players played 10 min. We collected external (GPS parameters) and internal (MB concentration and CK activity) loads after the match, and their correlations were analyzed. On the match day, the weather was sunny, with a temperature of 18.0 °C and humidity of 46%. The wind speed was 3.5 m/s, and the pitch was natural grass.

MB and CK measurements

Whole blood samples were collected from the participants by fingertip blood sampling before the match (baseline) and at 2, 16, and 40 h after the match. This was performed by sterilising the fingertip, sticking blood collection tape onto the fingertip, and puncturing the fingertip with a lancet (BD Microtainer Safety Lancet; Becton, Dickinson and Company, Tokyo, Japan) (Figure 1).

To measure MB concentration, the IA-100 compact immunoassay system developed by Sysmex Corporation was used.[11] Using microfluidic reaction technology, this test system can be used for highly sensitive chemiluminescent enzyme immunoassays. This system is a compact machine that can be used in a wide range of applications. The system consists of a DVD-size reagent cartridge with microfluidic channels and a device with a centrifugation mechanism, magnet, and photodetector (Figure 1). The small size of the device (approximately 17 cm^2) allows it to be placed in a variety of locations. A few drops (approximately 35 µL) of a heparinized whole blood sample were dropped into the reagent cartridge, and the centrifugation function (rotational speed of 3600 rpm) was used to separate the plasma and start the measurement. The device can be connected to the Internet and functions as an Internet of Things device that provides users with daily accuracy management and predictive failure information. The data management system used in this study was an application called the Lifee™ system (Micro Blood Sci-

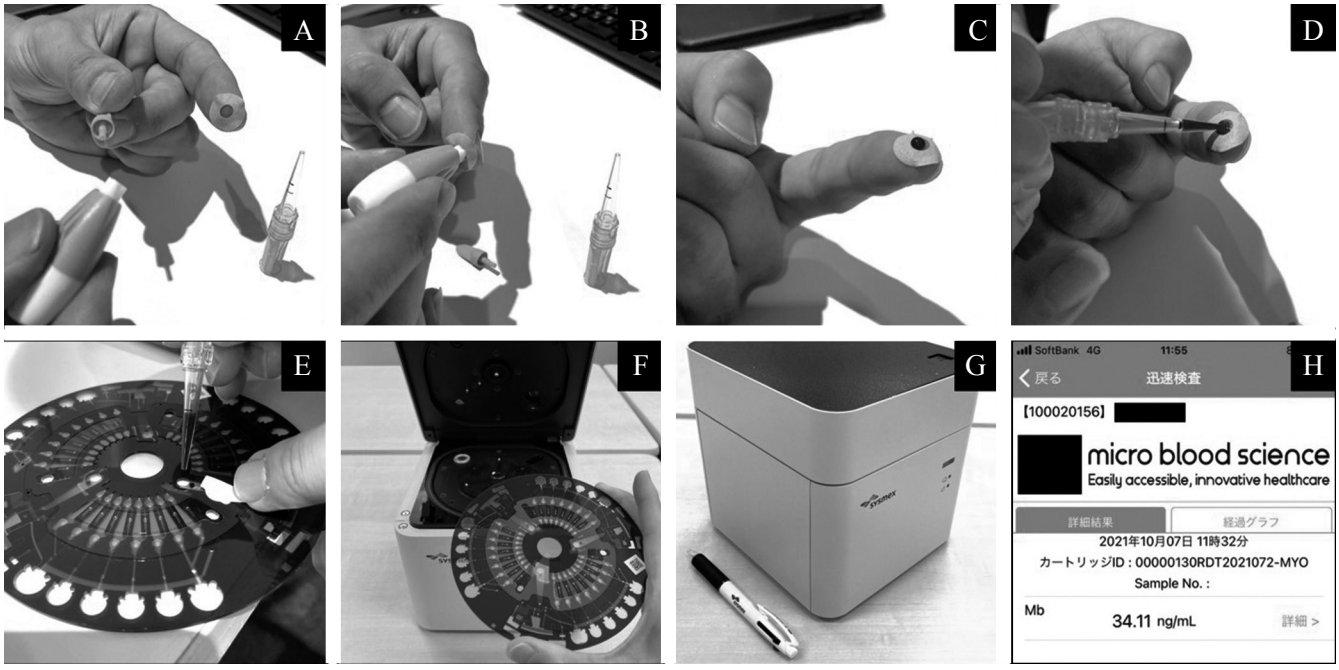

Figure 1.—Myoglobin measurement using the portable IA-100 immunoassay system. A-D) The method of fingertip blood sampling. The fingertip is pricked with a 21-gauge lancet after which blood is drawn by the capillary phenomenon. E) A volume of 35 µL of a whole blood sample is dropped into the reagent cartridge. F) The cartridge is put into the portable IA-100 system. G) Image of a centrifuge for spinning the blood sample. H: The data are sent to the athlete's smartphone application.

ence Inc., Chiyoda City, Tokyo, Japan), which is a data management system on the cloud. This system can register MB test orders and allow users to view the results using a smartphone application.

To measure CK concentration, 60 µL of whole blood was collected using an MBS EDTA-2K capillary (Micro Blood Science Inc.). All samples were stored at 4° C during their transfer to the clinical laboratory. In the clinical laboratory, plasma was separated by centrifugation at 1450×g for 5 min. A volume of 15 µL of plasma sample from the fingertip blood was then quantitatively diluted with 190 µL of physiological saline before measurement. The CK biochemical parameters were evaluated using a JCA-BM6010 automated biochemical analyzer (JEOL Ltd., Tokyo, Japan).

Measurement of player loads using a GPS device

Match loads were monitored using a GPS device (Vector X7; Catapult Sports, Melbourne, Australia).[12] After the match, GPS receivers were removed from the players and downloaded to a computer for analysis. The metrics derived from the device were the total distance (m), high-speed running distance (m) ([HSR] 19.8-24 km/h), sprint distance (>24 km/h), high acceleration count (n) (>3 m/s^2), high deceleration count (n) (<-3 m/s^2), and the sum of the combined triaxial accelerometer data (PlayerLoad [PL]).[13]

Statistical analysis

To determine whether there were significant changes in MB concentration and CK activity after the match, a one-way repeated measures of analysis of variance followed by the Bonferroni *post-hoc* test were used. Correlations between the external and internal load parameters were analyzed using partial correlation analysis normalized to baseline concentrations of MB and CK. Statistical analysis was performed using SPSS v. 26 (IBM Corp., Armonk, NY, USA). P values <5% were considered to be statistically significant.

Results

Post-match changes in MB and CK

The time course of changes in MB and CK concentrations is shown in Figure 2. The mean MB concentration at baseline was 38.3 ng/mL (95% confidence interval, 34.4-

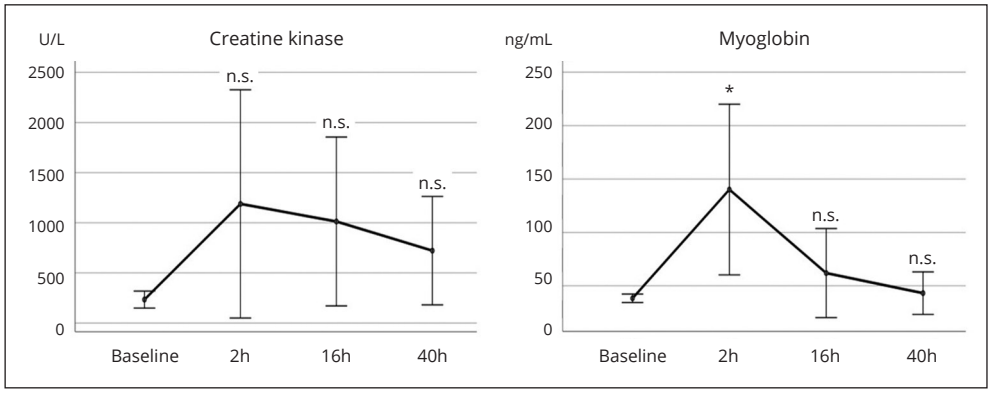

Figure 2.—Changes in MB and CK concentrations post-match.
n.s.: not significant.
*Significant difference *versus* all other time points (P<0.05).

42.3), whereas MB concentrations at 2, 16, and 40 h after the match were 140.1 (60.2-220.0), 61.9 (20.2-103.6), and 43.2 (23.2-63.1) ng/mL, respectively. We found that the mean MB concentration was significantly higher at 2 h than at the other time points ($P<.05$). Our results indicated that MB concentration was elevated 2 h after the match and decreased to baseline levels within 16 h post-match.

CK concentrations at baseline, 2, 16, and 40 h after the match were 234.3 (149.6-319.0), 1188.3 (50.4-2326.1), 1012.8 (171.0-1854.6), and 721 (181.3-1262.4) U/L, respectively. In contrast to our findings for MB concentrations, we found that, although the mean CK concentration was increased after the match compared with the baseline concentration, there were no significant differences in CK concentrations between any of the evaluated time points. There were substantial differences in CK concentrations between individuals. The elevated CK concentrations in this study did not recover even 40 h after the match.

External and internal loads

We analysed GPS data for the 15 field players after the match. The average total distance of full-time play (90 min) for five players was 10,826.4 m. The mean HSR was 485.6 m, the mean sprint distance was 254.6 m, and the mean total PL was 933.8. There were 62.0 accelerations and 66.8 decelerations.

Table I shows the correlations between GPS parameters (external load) and MB/CK concentrations (internal load), adjusted for baseline concentrations of MB and CK. MB concentration at 2 h was strongly correlated with total accelerations (r=0.87), total distance (r=0.82), total decelerations (r=0.81), and total PL (r=0.78), while HSR and sprint distance were not correlated with MB concentration. These observations suggested that acceleration/deceleration loaded more stress on the muscle than sprinting and HSR.

Table I.—*Correlations between GPS data and MB/CK levels.*

Parameters	Myoglobin			Creatine kinase		
	2 h	16 h	40 h	2 h	16 h	40 h
Total distance	0.82*	0.77*	0.72	0.71	0.71	0.69
High-speed running distance	0.66	0.63	0.59	0.54	0.53	0.51
Sprint distance	0.58	0.57	0.53	0.46	0.46	0.44
Total player load	0.78*	0.71	0.65	0.64	0.64	0.62
Accelerations	0.87*	0.82*	0.76*	0.77*	0.76*	0.75
Decelerations	0.81*	0.75	0.68	0.69	0.69	0.67

CK: creatine kinase; GPS: global positioning system; MB: myoglobin.
*Statistically significant correlation (P<0.01).

MB concentration at 16 h was strongly correlated with accelerations (r=0.82) and the total distance (r=0.77), while that at 40 h was correlated with accelerations (r=0.76). CK concentrations at 2 h and 16 h were only correlated with accelerations (r=0.77 and r=0.76, respectively), while CK concentration at 40 h was not correlated with any of the evaluated parameters.

Discussion

This study reveals that MB was a more sensitive marker of muscle damage than CK after football match-play in male professional players. Although several studies have already demonstrated a similar result to that in our study, the measurement of MB is not common in playing sports. The reason for this lack of measurement is a lack of appropriate devices to easily measure MB. In this study, we used a novel portable device to measure MB on-site. This device enables a more convenient measurement of MB concentration, providing real-time data on a player's muscle damage status. This system enables athletes to determine their daily MB concentration even if they are in their homes.

The knowledge of MB concentrations helps not only athletes but also physical coaches to monitor athletes' condition regarding whether they have recovered from a prior physical training load and are ready to progress to another physical training session. This knowledge could not only help personalize an athlete's training load to optimize their trainability, but also avoid sports injuries. Furthermore, this device could be used for sports other than football and could utilize other immunoassays that practitioners are interested in. Therefore, our findings could be applied in a broad context.

The development of wearable devices enables the monitoring of athletes' biological information. In addition, development in the field of the Internet of Medical Things (IoMT) has dramatically changed our strategies regarding monitoring and managing athletes. The IoMT can provide a large amount of data, but practitioners need to have a proper understanding of the data and make appropriate decisions. The misunderstanding of data can lead athletes to take an inadequate direction in training. A GPS is a wearable device that is widely used in various sports. This study showed that the number of accelerations/decelerations caused more burden to the muscle than the sprint/HSR distance. Russell *et al.*[14] performed similar observations in England Premier League under-21 football players. They reported that the sprint/HSR distance and the number of accelerations/decelerations were correlated with CK concentrations. These observations suggested that it is necessary to focus not only on sprinting/HSR but also on acceleration/deceleration parameters to monitor muscle damage in football players. Schuth *et al.*[15] investigated the relationship between external/internal load and CK concentrations using the GPS system in football players, and concluded that the relationship changes are position-specific and might depend on the type of session (training or match). Interestingly, they reported that morning CK concentrations also contributed to the next day's CK concentrations. Similarly, in our study, elevated CK concentrations lasted a few days, while elevated MB concentrations decreased by the next morning, suggesting that MB could become a more reliable muscle biomarker than CK in sports medicine. To the best of our knowledge, there have been no previous reports that investigated the relationship between GPS and MB data in professional football match-play.

Recovery after a competition is important to decrease the likelihood of muscle injury and maintain athletes' performance levels. Incomplete recovery might result in underperformance and injury, making the investigation of muscle damage markers important. Ruf *et al.*[16] assessed acute responses to load in high-level youth football players, and found limited responsiveness of athlete-reported questionnaires and countermovement jump parameters. Saidi *et al.*[9] reviewed previous publications and found eight studies that investigated the effects of football training on muscle damage markers. Most studies monitored CK followed by lactate dehydrogenase, and these parameters were elevated after training and increased for a long time, which suggested that they were not good indicators to assess short-term recovery from muscle damage. In our study, all players showed elevated MB concentration at 2 h. Three players continued to exhibit elevated MB concentration (more than 1.5-fold) at 16 h as compared to the baseline, and only one player continued to have elevated MB concentration at 40 h (Figure 2). Therefore, the measurement of MB could be a good indicator to screen players at risk of muscle damage.

The development of IoMT enables the performance of previously difficult innovative sports-related medical and physical investigations. This study showed previously unknown associations between external/internal load in football match-play using IoMT devices (GPS and IA-100). Acceleration and deceleration increased muscle damage during the match, and a prolonged increase in MB concentration two days after the match could be a risk for incomplete recovery from match load. This information could be useful not only for athletes but also for practitioners, to manage the training load or perform adequate recovery strategies.

Limitations of the study

There are some limitations to this study. First, participants belonged to only one male professional football club. Therefore, the generalizability of our findings is not guaranteed in other sports, ages, or sex. Second, there were only a small number of participants. Third, we did not evaluate physical performance data in this study. Therefore, whether increased MB concentrations were related to low physical performance is unknown. Further studies using IoMT devices may solve these limitations.

Conclusions

In conclusion, this study shows that MB concentration is a more sensitive biomarker of muscle damage after a football match than CK activity. Muscle damage during football match-play is strongly associated with accelera-

tion/deceleration metrics rather than the HSR distance. Monitoring MB in athletes can help them determine their recovery status from the prior training load and help practitioners to manage the training load.

References

1. Mikkelsen TS, Toft P. The correlation between creatine kinase and myoglobin in critical ill patients with rhabdomyolysis. Crit Care 2004;8(Suppl 1):155.

2. Hasić S, Jadrić R, Kiseljaković E, Radovanović J, Winterhalter-Jadrić M. Comparison of creatine kinase activity and myoglobin blood level in acute myocardial infarction patients. Bosn J Basic Med Sci 2006;6:19–23.

3. Lazarim FL, Antunes-Neto JM, da Silva FO, Nunes LA, Bassini-Cameron A, Cameron LC, *et al.* The upper values of plasma creatine kinase of professional soccer players during the Brazilian National Championship. J Sci Med Sport 2009;12:85–90.

4. Ordway GA, Garry DJ. Myoglobin: an essential hemoprotein in striated muscle. J Exp Biol 2004;207:3441–6.

5. Şenışık SÇ, Akova B, Şekir U, Gür H. Effects of muscle architecture on eccentric exercise induced muscle damage responses. J Sports Sci Med 2021;20:655–64.

6. Hettling H, van Beek JH. Analyzing the functional properties of the creatine kinase system with multiscale 'sloppy' modeling. PLOS Comput Biol 2011;7:e1002130.

7. Brancaccio P, Lippi G, Maffulli N. Biochemical markers of muscular damage. Clin Chem Lab Med 2010;48:757–67.

8. Ehrmann FE, Duncan CS, Sindhusake D, Franzsen WN, Greene DA. GPS and injury prevention in professional soccer. J Strength Cond Res 2016;30:360–7.

9. Saidi K, Abderrahman AB, Hackney AC, Bideau B, Zouita S, Granacher U, *et al.* Hematology, hormones, inflammation, and muscle damage in elite and professional soccer players: a systematic review with implications for exercise. Sports Med 2021;51:2607–27.

10. Nosek P, Brownlee TE, Drust B, Andrew M. Feedback of GPS training data within professional English soccer: a comparison of decision making and perceptions between coaches, players and performance staff. Sci Med Footb 2021;5:35–47.

11. Dupuy AM, Né M, Bargnoux AS, Badiou S, Cristol JP. Analytical evaluation of Lumipulse® BRAHMS PCT CLEIA assay and clinical performances in an unselected population as compared with central lab PCT assay. Clin Biochem 2017;50:248–50.

12. Theodoropoulos JS, Bettle J, Kosy JD. The use of GPS and inertial devices for player monitoring in team sports: A review of current and future applications. Orthop Rev (Pavia) 2020;12:7863.

13. Catapult Support. 2023 [Internet]. Available from: https://support.catapultsports.com/hc/en-us [cited 2023, Jul 26].

14. Russell M, Sparkes W, Northeast J, Cook CJ, Bracken RM, Kilduff LP. Relationships between match activities and peak power output and Creatine Kinase responses to professional reserve team soccer matchplay. Hum Mov Sci 2016;45:96–101.

15. Schuth G, Szigeti G, Dobreff G, Revisnyei P, Pasic A, Toka L, *et al.* Factors influencing creatine kinase response in youth national team soccer players. Sports Health 2021;13:332–40.

16. Ruf L, Drust B, Ehmann P, Skorski S, Meyer T. Are measurement instruments responsive to assess acute responses to load in high-level youth soccer players? Front Sports Act Living 2022;4:879858.

Conflicts of interest
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Authors' contributions
All authors contributed equally to the manuscript and read and approved the final version of the manuscript.

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